

ABET Self-Study Evaluation and Continuous Improvement

Course: CHE 311 – Transport Phenomena I, Fall 2025

1. Self-Study Evaluation (Findings and Interpretation)

The assessment of Student Outcome 1 (SO1) in CHE 311 was conducted using the comprehensive final exam as a direct measure of student ability to apply principles of momentum transport and fluid mechanics to solve complex engineering problems.

The **average exam score of 70.97** indicates that, overall, students demonstrated an acceptable level of competency in core topics such as fluid statics, conservation of momentum, dimensional analysis, and internal/external flow. However, the **standard deviation of 17.19** suggests a notable spread in performance, indicating inconsistent levels of mastery across the cohort.

The **performance indicator average of 2.56/3.0** exceeds the program's attainment benchmark (2.0), confirming that **SO1 is achieved** for this course. The distribution of student performance further supports this conclusion, with **(21/36)** performing at or above the developing level and **demonstrating satisfactory proficiency**.

A subset of students struggles with fundamental concepts and/or the integration of multiple principles required for solving complex fluid mechanics problems. In particular, variability in performance may be attributed to:

- Difficulty in translating physical systems into mathematical models
- Weaknesses in prerequisite knowledge (e.g., differential equations)
- Challenges in multi-step problem-solving under exam conditions

Additionally, the relatively high variability in exam scores indicates that while high-performing students excel, others are not consistently reaching the expected level of competency.

2. Continuous Improvement Actions

Based on the assessment findings, the following targeted actions will be implemented to improve student learning and reduce performance variability:

A. Strengthening Conceptual Foundations

- Introduce **short, focused concept reviews** (e.g., mini-lectures or pre-class modules) on key prerequisite topics such as force balances, units/dimensions, and differential analysis.
- Incorporate **concept-check questions** during lectures to reinforce understanding of fundamental fluid mechanics principles (e.g., pressure variation, shear stress, velocity profiles).

B. Enhancing Problem-Solving Skills

- Provide **structured problem-solving frameworks** (step-by-step methodologies) for complex topics such as Navier–Stokes simplifications and momentum balances.
- Integrate **worked examples with increasing complexity**, emphasizing problem setup, assumptions, and interpretation of results.
- Use **active learning strategies** (e.g., group problem-solving sessions) to improve student engagement and analytical thinking.

C. Early Identification and Support for At-Risk Students

- Implement **early-semester formative assessments** to identify students struggling with core concepts.
- Offer **targeted review sessions, tutoring, or supplemental instruction** focused on high-risk topics.
- Encourage use of office hours and peer-assisted learning opportunities.

D. Improving Assessment Design and Feedback

- Conduct a **post-exam item analysis** to identify specific topics or question types with high error rates (e.g., internal flow, boundary layer concepts).
- Adjust future assessments to ensure **alignment between learning objectives, instruction, and evaluation**.
- Provide **timely and detailed feedback** on exams and assignments to help students understand mistakes and improve.

E. Increasing Continuous Assessment Opportunities

- Incorporate **low-stakes quizzes and homework assignments** throughout the semester to reinforce learning and reduce reliance on high-stakes exams.

3. Reassessment Plan

The effectiveness of these continuous improvement actions will be evaluated in the next offering of CHE 311 by:

- Comparing **performance indicator averages** and **distribution of S/D/U ratings**
- Monitoring changes in **standard deviation** to assess reduction in performance variability
- Reviewing **student performance on targeted topics** identified in the current assessment

4. Conclusion

Student Outcome 1 is **attained** in CHE 311; however, the assessment reveals **opportunities for improvement in consistency and support for lower-performing students**. The planned actions focus on strengthening foundational understanding, enhancing structured problem-solving skills, and providing earlier and more targeted interventions. These efforts are expected to improve overall student performance and reduce the proportion of unsatisfactory outcomes in future course offerings.

